EKF for Spacecraft Navigation

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Abstract

This paper presents the development of an extended Kalman filter (EKF) for spacecraft navigation using global positioning system (GPS) measurements. The EKF is formulated by combining a two-body orbital dynamics model with GPS pseudorange measurements from multiple satellites. The dynamical model propagates the spacecraft state vector, i.e., inertial position and velocity along with clock bias, while the update step corrects the estimate using incoming GPS measurements. Formulation of the measurement model accommodates for measurements from a varying number of GPS satellites in view. The performance of the navigation filter is evaluated through simulation using satellite ephemeris data and raw GPS measurements.

1 Introduction

The EKF is a widely used nonlinear estimation technique that enables recursive state estimation in systems with nonlinear dynamics and measurement models. By linearizing the system about the current state estimate, the EKF propagates the spacecraft state forward in time using a dynamical model and then corrects the estimate using incoming GPS measurements [1]. This framework allows the filter to reduce the impact of measurement noise while maintaining an accurate estimate of the spacecraft trajectory.

Vigneron in [2] presents development of a GPS measurement model combined with models for dynamics and perturbations. This is accompanied with different Kalman filtering techniques to carry out autonomous spacecraft navigation. Following similar framework, this work formulates a spacecraft navigation filter by combining a two-body orbital dynamics model with GPS pseudorange measurements from multiple satellites. The EKF is implemented to estimate the spacecraft position, velocity, and receiver clock bias. The performance of the navigation filter is evaluated through simulation using GPS satellite ephemeris data from an `Sp3` file, and GPS pseudorange measurements from a `RINEX` file. Section 2 presents development of the dynamical and measurement models, followed by the filter formulation. Section 3 discusses the simulation results, while Sec. 4 presents the concluding remarks.

2 Navigation Filter

This section presents development of the dynamics and measurement models, and the filter formulation for EKF state estimation. For completeness, the two reference frames that are used in this development are the Earth centered inertial reference frame ($\mathcal{F}_I$), and Earth centered Earth fixed reference frame ($\mathcal{F}_F$) as defined by De Ruiter et al. in [3].

2.1 Dynamics Model

The dynamics of the spacecraft-Earth system is based on the two body problem, used to propagate the position and velocity of the spacecraft in $\mathcal{F}_I$. Mathematically, this is

$$\ddot{\mathbf{r}} = \frac{-\mu_\oplus}{r^3}\mathbf{r} \tag{1}$$

where $\mathbf{r}$ is the position vector of the spacecraft, $\ddot{\mathbf{r}}$ is the acceleration vector, $r = \|\mathbf{r}\|$ is the magnitude of the position vector, and $\mu_\oplus$ is the gravitation parameter for Earth. Since the value of $\ddot{\mathbf{r}}$ is dependent on r, the system cannot be modelled as a linear system as shown below:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B} \tag{2}$$

Therefore, to linearize the system, Jacobian of the dynamics function must be evaluated. For a state vector given by $\mathbf{x} = \begin{bmatrix} x & y & z & \dot{x} & \dot{y} & \dot{z} & b_{clk} \end{bmatrix}^\top$, the dynamics function is:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \int_{t_k}^{t_{k+1}} f(\mathbf{x}(t), t)\, \mathrm{d}t \tag{3}$$

$$f(\mathbf{x}(t), t) = \begin{bmatrix} \mathbf{v}(t) \\ -\big(\mu_\oplus / r(t)^3\big)\mathbf{r}(t) \\ 0 \end{bmatrix} \tag{4}$$

Readers must note that the drift on clock bias b_{clk} is not captured by this dynamics model. Therefore, $\dot{b}_{clk} = 0$.

Next, the Jacobian matrix $\mathbf{F}$ is found as

$$\mathbf{F}(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} f(\mathbf{x}(t), t) = \begin{bmatrix} \partial\mathbf{v}/\partial\mathbf{r} & \partial\mathbf{v}/\partial\mathbf{v} & \partial\mathbf{v}/\partial b_{clk} \\ \partial\ddot{\mathbf{r}}/\partial\mathbf{r} & \partial\ddot{\mathbf{r}}/\partial\mathbf{v} & \partial\ddot{\mathbf{r}}/\partial b_{clk} \\ \partial\dot{b}_{clk}/\partial\mathbf{r} & \partial\dot{b}_{clk}/\partial\mathbf{v} & \partial\dot{b}_{clk}/\partial b_{clk} \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{3\times3} & \mathbf{I}_{3\times3} & \mathbf{0}_{3\times1} \\ \mu_\oplus\big(\frac{3\mathbf{r}\mathbf{r}^\top}{r^5} - \frac{\mathbf{I}_{3\times3}}{r^3}\big) & \mathbf{0}_{3\times3} & \mathbf{0}_{3\times1} \\ \mathbf{0}_{1\times3} & \mathbf{0}_{1\times3} & 0 \end{bmatrix} \tag{5}$$

Finally, the state transition matrix is given by:

$$\mathbf{\Phi} = \exp(\mathbf{F}\, \mathrm{d}t) \tag{6}$$

where $\exp(\mathbf{F}\, \mathrm{d}t)$ can be computed by employing Taylor series expansion. This yields:

$$\mathbf{\Phi} = \exp(\mathbf{F}\, \mathrm{d}t) = \mathbf{I}_{7\times7} + \mathbf{F}\, \mathrm{d}t + \frac{(\mathbf{F}\, \mathrm{d}t)^2}{2!} + \frac{(\mathbf{F}\, \mathrm{d}t)^3}{3!} + \dots \tag{7}$$

2.2 Measurement Model

This study makes use of the onboard GPS measurements to update the filter prediction. Time taken by the GPS signal to travel from the transmitter to the receiver is used to calculate distance between the two satellites. This distance is called the pseudorange. Meanwhile, the dynamics model propagates the spacecraft state vector, i.e. inertial position and velocity along with clock bias. Therefore, a measurement model must be developed in order to compare the predicted and measured states. This is achieved by calculating the pseudorange from the predicted states as shown below:

$$h(\mathbf{x}) = \|\underbrace{\mathbf{C}_{FI}\mathbf{r} - \mathbf{r}_{sv}}_{\mathbf{d}}\| - c(\delta t_{sv} - b_{clk}) \tag{8}$$

where $\mathbf{C}_{FI}$ is a principle rotation about z axis by an angle θ_{GMT}, $\mathbf{r}_{sv}$ is the position vector of the GPS satellite in $\mathcal{F}_F$, c is the speed of light and δt is the time correction, and $\mathbf{d}$ is the line of sight vector such that its norm gives the pseudorange ρ. Since this measurement model consists of nonlinearities, a Jacobian of this measurement matrix $h(\mathbf{x})$ must be computed. This Jacobian matrix $\mathbf{H}(\mathbf{x})$ is given by:

$$\mathbf{H}(\mathbf{x}) = \frac{\partial h(\mathbf{x})}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial h(\mathbf{x})}{\partial \mathbf{r}} & \frac{\partial h(\mathbf{x})}{\partial \mathbf{v}} & \frac{\partial h(\mathbf{x})}{\partial b_{clk}} \end{bmatrix} \tag{9}$$

$$= \begin{bmatrix} \frac{\mathbf{d}^\top}{\rho}\mathbf{C}_{FI} & \mathbf{0}_{1\times3} & c \end{bmatrix} \tag{10}$$

Since the spacecraft at any given instant has n GPS satellites in view, there are n pseudorange measurements recorded at each time instant. Therefore, a measurement vector $\mathbf{h}(\mathbf{x})_{n\times1}$ is defined to map the propagated states to the pseudorange measurement. This vector is given by:

$$\mathbf{h}(\mathbf{x}) = \begin{bmatrix} h_1(\mathbf{x}) \\ h_2(\mathbf{x}) \\ \vdots \\ h_n(\mathbf{x}) \end{bmatrix} \tag{11}$$

such that

$$h_i(\mathbf{x}) = \|\mathbf{C}_{FI}\mathbf{r} - \mathbf{r}_{sv_i}\| - c(\delta t_{sv_i} - b_{clk}) \qquad \forall i = 1, 2, \dots n \tag{12}$$

Consequently, a measurement Jacobian vector is also defined, given by:

$$\boldsymbol{\mathcal{H}}(\mathbf{x}) = \begin{bmatrix} \mathbf{H}_1(\mathbf{x}) \\ \mathbf{H}_2(\mathbf{x}) \\ \vdots \\ \mathbf{H}_n(\mathbf{x}) \end{bmatrix} \tag{13}$$

such that

$$\mathbf{H}_i(\mathbf{x}) = \begin{bmatrix} \frac{\mathbf{d}_i^\top}{\rho_i}\mathbf{C}_{FI} & \mathbf{0}_{1\times 3} & c \end{bmatrix} \qquad \forall i = 1, 2, \ldots n \tag{14}$$

2.3 Filter Formulation

This section presents formulation of the EKF. The design of this EKF is based on the formulation presented by Carpenter and D'Souza in [4]. Propagation step uses previous step's state and covariance estimates to predict the next step's states and covariance. This is done using the dynamics model discussed in Sec. 2.1:

$$\hat{\mathbf{x}}_{k+1}^- = \hat{\mathbf{x}}_k^+ + \int_{t_k}^{t_{k+1}} f(\hat{\mathbf{x}}_k^+(t), t)\, \mathrm{d}t \tag{15}$$

$$\mathbf{P}_{k+1}^- = \boldsymbol{\Phi}_k \mathbf{P}_k^+ \boldsymbol{\Phi}_k^\top + \mathbf{Q}_k \tag{16}$$

where subscripts k and $k+1$ denote the previous and current time steps respectively, superscripts $-$ and $+$ denote the priori and posteriori estimates respectively, the hat $\hat{()}$ denotes estimated state vector, and $\mathbf{Q}_k$ is the process noise covariance.

The correction step compares the priori estimate state vector with GPS measurements to find the innovation $\boldsymbol{\mathcal{I}}_k$. This is followed by calculation of the Kalman gain $\mathbf{K}_k$, using the measurement Jacobian $\boldsymbol{\mathcal{H}}(\hat{\mathbf{x}}_k)$ as:

$$\boldsymbol{\mathcal{I}}_k = \mathbf{y}_k - \mathbf{h}(\hat{\mathbf{x}}_{k+1}^-) \tag{17}$$

$$\mathbf{K}_k = \mathbf{P}_{k+1}^- \boldsymbol{\mathcal{H}}(\hat{\mathbf{x}}_{k+1}^-)^\top \left(\boldsymbol{\mathcal{H}}(\hat{\mathbf{x}}_{k+1}^-) \mathbf{P}_{k+1}^- \boldsymbol{\mathcal{H}}(\hat{\mathbf{x}}_{k+1}^-)^\top + \mathbf{R} \right)^{-1} \tag{18}$$

where $\mathbf{y}_k$ is the GPS measurement vector and $\mathbf{R}$ is the measurement noise covariance. Finally, the posteriori state vector and covariance estimates are computed using this innovation and Kalman gain:

$$\hat{\mathbf{x}}_{k+1}^+ = \hat{\mathbf{x}}_{k+1}^- + \mathbf{K}_k \boldsymbol{\mathcal{I}}_k \tag{19}$$

$$\mathbf{P}_{k+1}^+ = \left(\mathbf{I}_{7\times 7} - \mathbf{K}_k \boldsymbol{\mathcal{H}}(\hat{\mathbf{x}}_{k+1}^-) \right) \mathbf{P}_{k+1}^- \left(\mathbf{I}_{7\times 7} - \mathbf{K}_k \boldsymbol{\mathcal{H}}(\hat{\mathbf{x}}_{k+1}^-) \right)^\top + \mathbf{K}_k \mathbf{R} \mathbf{K}_k^\top \tag{20}$$

3 Simulation Results

This section presents the results of the simulation run to validate the EKF formulation presented in Sec. 2.3. The simulation was designed to read an `Sp3` file containing precise information of GPS satellites. The clock values along with x, y, and z positions of the GPS satellites were extracted. This was followed by reading the GPS satellite IDs and their corresponding pseudorange values, with their respective position vectors. This was then interpolated to get values for spacecraft Epoch times. Table 1 displays the time and interpolation parameters used.

Table 1: Epoch time and interpolation parameters.

Parameter	Value
Time step between measurement epoch [s]	10
Number of epoch measurements	11
First epoch measurement [s]	0
Position interpolation polynomial order	11
Clock interpolation polynomial order	3
`Sp3` time grid relative to first epoch [mins]	(-78:15:73)

A comparison between EKF estimates and ground truth can be viewed on the left side of Fig. 1, while the right side shows position error between ground truth and estimated values.

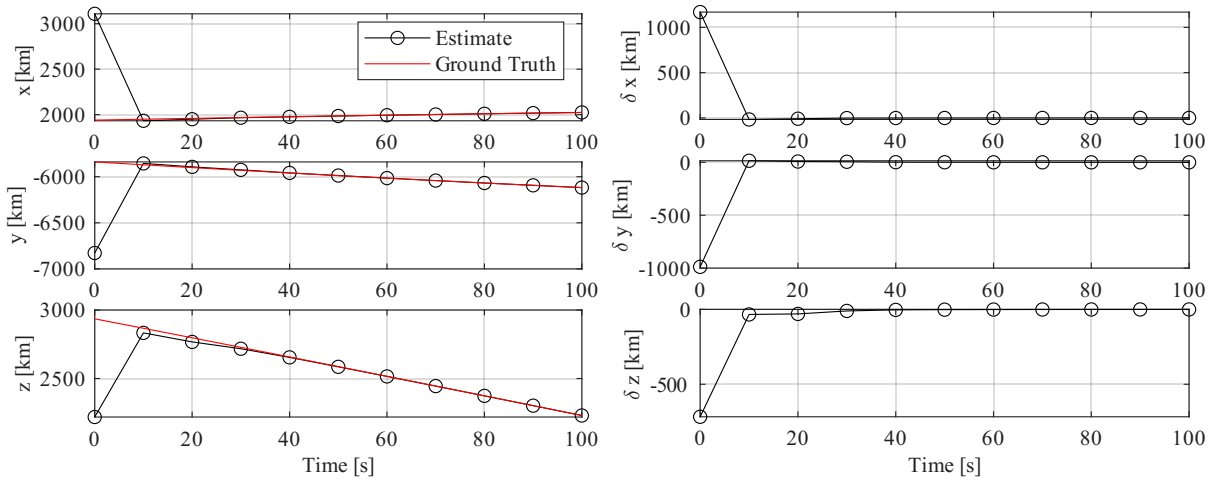

Figure 1: EKF estimates vs. ground truth values (left) and estimate error (right).

It is clear from the left side of Fig. 1 that the estimated position closely follows the ground truth trajectory. Although a huge error is observed initially, the estimated state is able to capture the overall motion of the spacecraft. The right side of Fig. 1 shows convergence of the error evolution. The error values for x, y, and z components settle at 36.75 m, 98.1 m, and 154.2 m respectively. Same trend is observed where the initial measurement has a large error. This error is a result of the chosen filter parameters. These filter parameters can be viewed in Table 2.

Table 2: Filter tuning parameters.

Parameter	Value
Initial Position Std. Dev.	10 km
Initial Velocity Std. Dev.	100 $\mathrm{m/s}$
Process Noise (Position)	16e04 m^2
Process Noise (Velocity)	67.24 $\mathrm{m}^2/\mathrm{s}^2$
Measurement Noise Std. Dev. (uncorrelated)	5 m

Note how the initial position standard deviation value in Table 2 is 10 km or 10,000 m. This tells the filter that position components of the initialization has an error. This is exactly is what is observed in Fig. 1 where the initial value has a large error. Single update at the second epoch significantly drops down this error demonstrating the effectiveness of this EKF. Figure 2 shows the covariance evolution plot on the left side, and the estimated vs. ground truth altitude plot on the right side.

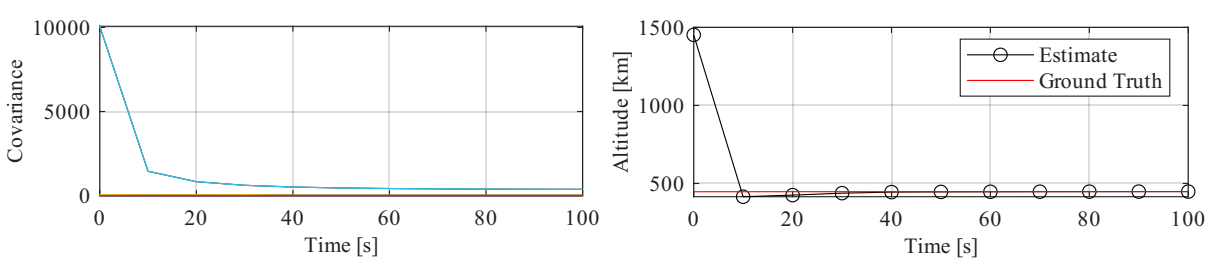

Figure 2: Covariance evolution (left) and estimated vs. ground truth altitude.

The covariance evolution extends the previously inferred observation of convergence. Furthermore, the plot also reveals

that the initial covariance for position is 10,000 which matches the tuning chosen in Table 2. Finally, the plot on the right side of Fig. 2 presents the evolution of the altitude estimation. It is clear that the altitude calculated using filter estimated state vectors settles close to the ground truth value after the fifth epoch.

4 Conclusion

To summarize, an EKF formulation is presented and tested in simulation. The simulation was designed to read an `Sp3` file to extract GPS satellites' ephemeris values. A `RINEX` file was read to extract IDs of GPS satellites in view and their respective pseudorange measurements along with the receiver clock values. The GPS position and clock values of corresponding satellite IDs were interpolated and used in the measurement model. The proposed filter formulation was used to estimate the satellite's position and velocity vectors. These estimates were compared with ground truth values. Results demonstrate that the EKF is effective and both the error and the covariance evolution show convergence.

References

[1] J. L. Crassidis and J. L. Junkins, *Optimal Estimation of Dynamic Systems*. Boca Raton, FL: CRC Press, second ed., 2011.

[2] A. C. Vigneron, "Nonlinear filtering for autonomous navigation of spacecraft in highly elliptical orbit," master of applied science, Carleton University, Ottawa, Ontario, Canada, May 2014.

[3] A. De Ruiter, C. Damaren, and J. Forbes, *Spacecraft Dynamics and Control: An Introduction*. Wiley, 2013.

[4] J. R. Carpenter and C. N. D'Souza, "Navigation filter best practices," Tech. Rep. NASA/TP-2018-219822, NASA Goddard Space Flight Center, April 2018.